

An **IP (Internet Protocol) address** is a unique numerical address assigned to every device connected to a network. It allows devices such as computers, smartphones, printers, and servers to communicate with each other over the Internet or a local network.

In **IPv4**, IP addresses are **32 bits long** and are divided into **five classes: A, B, C, D, and E**. Each class was designed for a different purpose based on the size of the network and the number of devices it needs to support.

IP Class	IP Address Range	Default Subnet Mask	Purpose	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	Used for very large networks because it supports millions of host devices.	Google or Amazon AWS data centers, where thousands or millions of servers and devices communicate.
Class B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	Used for medium-sized organizations such as universities and large companies.	A university campus network connecting classrooms, laboratories, libraries, and administrative offices.
Class C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	Used for small businesses and home networks with fewer devices.	A home Wi-Fi network connecting laptops, smartphones, printers, smart TVs, and security cameras.
Class D	224.0.0.0 – 239.255.255.255	Not Applicable	Reserved for Multicast communication , where data is sent to multiple devices simultaneously.	Live video streaming , IPTV, online webinars, or video conferencing services that send one stream to many users.
Class E	240.0.0.0 – 255.255.255.255	Not Applicable	Reserved for research, testing, and experimental purposes . It is not used for normal public networking.	Used by network researchers and developers for testing new networking technologies and experimental protocols.